SMART INDIA HACKATHON 2025

HAZARD WATCH

Problem Statement ID - 25039

Theme- Disaster Management

PS Category- Software

Team ID- 85557

Team Name- Coastal Sentinels

Problem Statement Title- Integrated Platform for Crowdsourced Ocean Hazard Reporting and Social Media Analytics

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HAZARD WATCH

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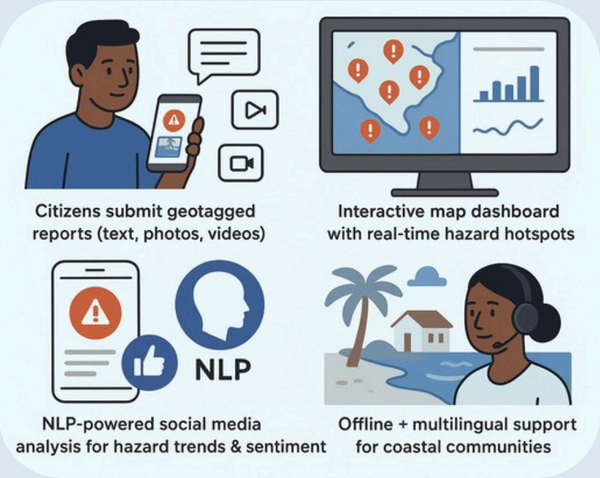
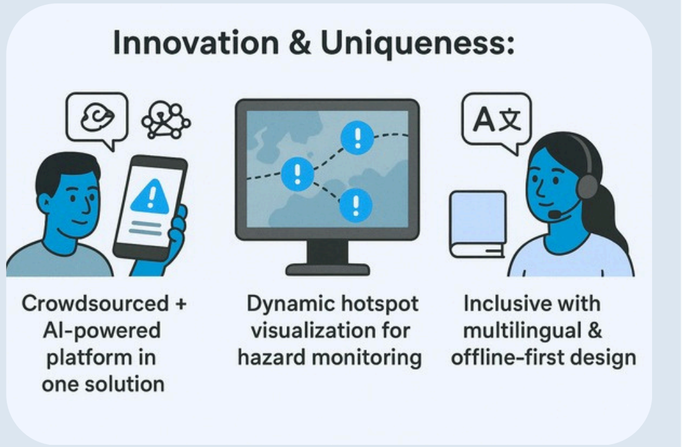
Proposed Solution



How it Addresses the Problem

Innovation & Uniqueness



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TECHNICAL APPROACH

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FUNCTIONALITY	TECHNOLOGY USED
Registration & Roles	React/Flutter + Firebase Auth
Hazard Alerts	Mapbox API, Node.js/Express
Incident Reporting	Firebase Storage, MongoDB
Hotspot/Map Routing	Mapbox, DBSCAN clustering
Team Coordination	React Dashboard + REST APIs
Sentiment Analysis	Python VADER/TextBlob, Twitter API
Multilingual/Offline	i18n, Service Workers

FLOW CHART OF FUNCTIONALITY

Citizens and Volunteers
Submit geotagged reports , photos and videos.

→

Backend servers & Databases
Storeand validate incoming reports.

→

Social Media NLP engiine
Extract hazard trends and public sentiment.

↓

Interactive Map Dashboard
Visualize hotspotand provide live hotspot.

←

INCOIS and Authorities
Enable faster decisions and response.

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IMPACT AND CHALLENGES

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FEASIBILITY

- **Built with readily available technologies** (mobile app, web, cloud, NLP).
- **Uses existing smartphones & internet → minimal extra infrastructure.**
- **Can be scaled gradually** (start with pilot in a few coastal districts).

POTENTIAL CHALLENGES

- **Ensuring data authenticity** (fake/misleading reports).
- **Low connectivity in remote coastal areas.**

IMPACT

- **Faster disaster response through real-time citizen reports + social media insights.**
- **Bridges data gap between satellites/sensors and ground-level conditions.**
- **Improves community resilience by involving citizens as active contributors.**

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BENEFITS AND PROTOTYPE

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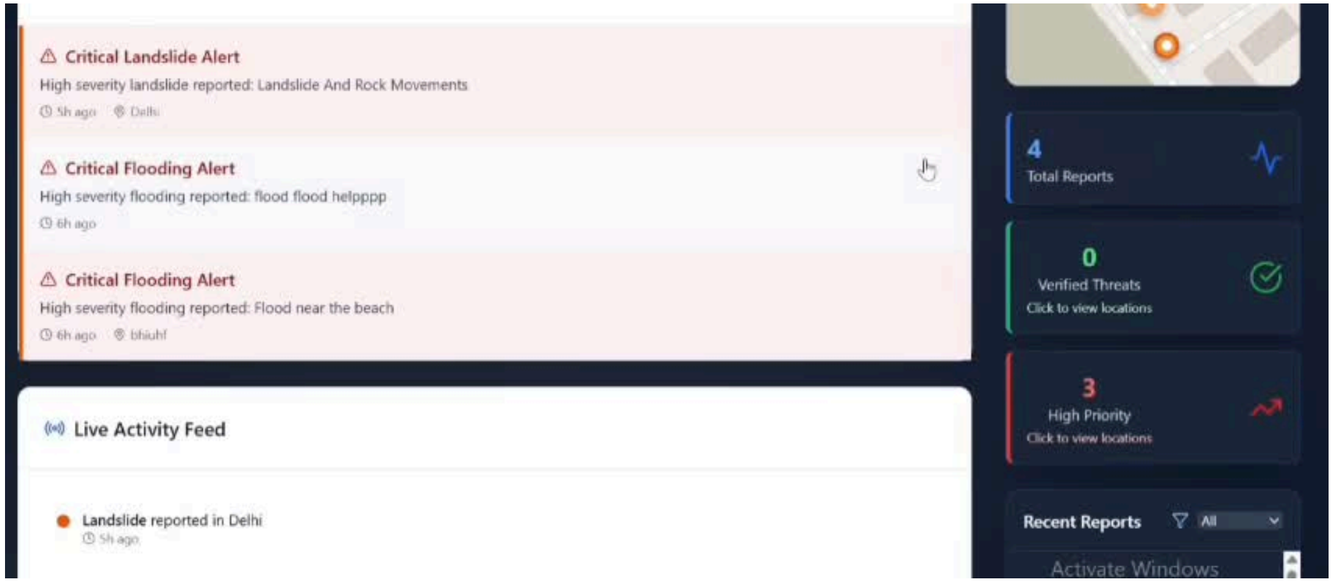
BENEFITS

- **Citizens: Quick alerts, safe route suggestions, and participation in coastal safety.**
- **Authorities** (INCOIS, disaster agencies)**: Better situational awareness & decision-making.**
- **Volunteers/Relief Teams: Coordinated response with live hazard maps.**
- **National Level: Strengthens disaster preparedness, supports Blue Economy goals, and aligns with Digital India & NEP 2020 for tech- driven governance.**

PROTOTYPE



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RESEARCH AND REFERENCES

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https://incois.gov.in/ 2004 Tsunami .

2012 Indian Ocean Tsunami Warning .
IOWave23 Drill (2023) .